

CS201 Midterm 2

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Abstract—Write only ONE function per question. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures.

I. QUESTION (BINARY SEARCH TREE), 20 POINTS

Write a recursive method in `TreeNode` class

```
bool greaterThanChildren()
```

which returns the number of nodes which value is greater than its children's average. You may assume left and right children must exist for all non-leaf nodes.

II. QUESTION (BINARY SEARCH TREE), 15 POINTS

Write a non-recursive method

```
int countOddNodes()
```

in `Tree` class that computes and returns the number of nodes with odd values in a binary search tree by using queue. You are only allowed to traverse the tree once. You may assume tree has at least 1 node.

III. QUESTION (QUEUE), 20 POINTS

Write static method in Array-based Queue implementation

```
Queue QueueZigZag(Queue[] list)
```

which constructs and returns a new array based queue by adding the elements in the *list* of queues one by one in zig-zag fashion. So, the first *k* elements of the original queue will be constructed with the first elements of the *k* queues in the list; the second *k* elements of the original queue will be constructed with the last elements of the *k* queues in the list, third *k* elements of the original

queue will be constructed with the second elements of the *k* queues in the list; etc. The elements from queues should be recreated (not copied from the queues). You are not allowed to use `enqueue`, `dequeue`, `isEmpty` functions. You should solve the question for array implementation. You may assume the length of each queue is same and even.

Below is an example:

Queue 1: 2 3 4 5

Queue 2: 1 7 9 2

Queue 3: 4 5 7 6

Resulting Queue: 2 1 4 5 2 6 3 7 5 4 9 7

IV. QUESTION (QUEUE), 15 POINTS

Write the method

```
int secondMaximum()
```

returns the second maximum number in a queue. Write the method for the linked list implementation.

V. QUESTION (HASHING), 15 POINTS

Write a static method in the Hash class (array implementation)

```
boolean equalPairDiffs(int[] array)
```

This method should return true if any two distinct pairs absolute difference in the array have the same value. Your method should run in $O(N^2)$ time. You may assume array has at least 2 items. You can use at most one external hash.

VI. QUESTION (HASHING), 15 POINTS

Write the static method in Hash class (Linked List implementation)

```
boolean sumOfThreeK(int[] array, int K)
```

that takes an array of integers as a parameter and returns true if the sum of three elements in the array is *K*. Your method should run in $O(N^2)$ time. Do not use any external data structures or arrays except the external hash table.